# Fatemeh Naeinian

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## Research Summary

Ph.D. student at New York University working on self-supervised representation learning and generalization under distribution shift, with autonomous driving as the main testbed. My research studies how visual pretraining affects the robustness of end-to-end driving models deployed in unseen cities, and develops JEPA-style pretraining and data-efficient adaptation methods for self-supervised encoders. My work spans large-scale benchmarking of self-supervised and supervised backbones in open- and closed-loop driving evaluation (nuScenes, NAVSIM), retrieval-augmented pretraining, and parameter-efficient fine-tuning.

**Interests:** Self-Supervised Learning · Robust Generalization · End-to-End Autonomous Driving · Data-Efficient Learning

## Education

**New York University** **Jan 2025 – Present**
*Ph.D. in Electrical Engineering, Expected Dec 2029* *New York, NY*
- GPA: 3.889/4.00

**University of Tehran** **Sep 2019 – Jun 2024**
*B.Sc. in Electrical Engineering (Control); Minor in Computer Engineering* *Tehran, Iran*
- Major GPA: 18.61/20.00; Minor GPA: 18.90/20.00

## Publications

- **F. Naeinian**, A. Hamza, H. Zhu, and A. Choromanska, "Zero-Shot Cross-City Generalization in End-to-End Autonomous Driving: Self-Supervised versus Supervised Representations." *arXiv:2603.11417*, 2026.
- **F. Naeinian**, E. Balazadeh, and M. Tale Masouleh, "Mapping Human Grasping to 3-Finger Grippers: A Deep Learning Perspective." In *Proceedings of the 32nd International Conference on Electrical Engineering (ICEE)*, pp. 1–7, 2024.

## Research Experience

**Graduate Research Assistant,** Learning Systems Laboratory **New York University**
*Advisor: Prof. Anna Choromanska* *Jan 2025 – Present*

- **Retrieval-Augmented JEPA for Generalizable Representations** (ongoing): Augmenting I-JEPA pretraining with retrieved support examples to test whether relational information between observations improves transfer under distribution shift; ablating retrieval quality, fusion mechanisms, and training objectives on controlled visual benchmarks (ImageNet-100).
- **Zero-Shot Cross-City Generalization in End-to-End Autonomous Driving** [arXiv]: Designed a geographic-generalization benchmark with strict city-based splits on nuScenes (open-loop) and NAVSIM (closed-loop, four cities).
- Pretrained ViT-S/14 encoders with I-JEPA, DINOv2, and MAE on driving data and integrated them into the LAW (camera-only) and TransFuser (camera + LiDAR) planners; benchmarked against supervised ResNet and Swin backbones across large multi-GPU training sweeps on SLURM clusters.
- Built scenario-level diagnostics (traffic density, stopping, turning) and a density-matched control experiment to attribute cross-city planning failures, revealing strongly directional generalization gaps.
- **Data-Efficient Adaptation of Self-Supervised Encoders** (ongoing): Developing data-selection methods that combine diversity sampling, prototype-based filtering, and gradient-based selection; comparing frozen, partially fine-tuned, and LoRA-adapted encoders on BDD100K semantic segmentation and object detection under limited labels.
- **TALKtoME: Educational Materials for Speech and Language Acquisition in Autism** [Project Website]: Designed data-collection strategies and curated real-world images and videos of action verbs and verb–noun pairs for a publicly released autism-focused language-learning dataset.

**Undergraduate Researcher,** TaarLab **University of Tehran**
*Advisor: Prof. Mehdi Tale Masouleh* *Jul 2023 – Mar 2024*

- Built a MediaPipe-based human hand-keypoint dataset and trained deep object-detection models to map human grasps to grasp points for a three-finger robotic gripper; published at ICEE 2024.

**Research Intern** **Technical University of Munich (Remote)**
*Supervisor: Dr.-Ing. Abdalla Swikir* *Summer 2022*

- Analyzed and compared computational methods for estimating the reachable-set boundaries of robotic manipulators.

## Technical Skills

- **Programming:** Python, C, C++, MATLAB, Bash
- **Machine Learning and Data:** PyTorch, torchvision, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn; PyTorch DDP, LoRA, Vision Transformers
- **Autonomous Driving:** nuScenes, NAVSIM, MMDetection3D/MMCV, BDD100K
- **Research Computing and Tools:** Linux, SLURM, Git, GitHub, Weights & Biases, Jupyter, Google Colab, LaTeX

## Poster Presentation

- "Zero-Shot Cross-City Generalization in End-to-End Autonomous Driving: Self-Supervised versus Supervised Representations." Poster presented at Computer Vision Day, New York, NY, Apr 2026.

## Leadership & Mentorship

- **Research Mentor, NYU ARISE Program**, New York University Summers 2025–2026
  Mentored high school students through end-to-end machine learning research, including problem formulation, dataset preparation, model training, experimental evaluation, and result interpretation. Developed introductory Python and machine learning instruction and coached students in scientific writing, poster design, and research presentation.

## Academic Service

- Peer Reviewer, *IEEE Robotics and Automation Letters (RA-L)*, 2026.
- Peer Reviewer, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

## Selected Graduate Coursework

- Advanced Machine Learning; Natural Language Processing; Reinforcement Learning; Computer Vision; Deep Learning; Applied Matrix Theory; Probability and Stochastic Processes; Digital Signal Processing.

## Honors and Awards

- School of Engineering Ph.D. Fellowship, New York University, 2025.
- Ranked 3rd in the Control Engineering track, Department of Electrical Engineering, University of Tehran, 2023.
- Academic Excellence Award, Supporter Foundation of the University of Tehran, 2019–2021.
- Ranked in the top 0.1% nationally in two fields of Iran's University Entrance Examination: 92nd in Mathematics and Physics and 91st in Foreign Languages, 2019.

## Teaching Experience

- **Course Assistant, New York University**: Digital Signal Processing, Fall 2025.
- **Teaching Assistant, University of Tehran**: Neural Networks and Deep Learning, Engineering Mathematics, Linear Control Systems, Probability and Statistics, Operations Research, and Linear Algebra, 2022–2023.

## Languages

- English (fluent); Persian (native).